Analytical Geometry & Calculus I (MTH111)

Nile University

Lecture #1

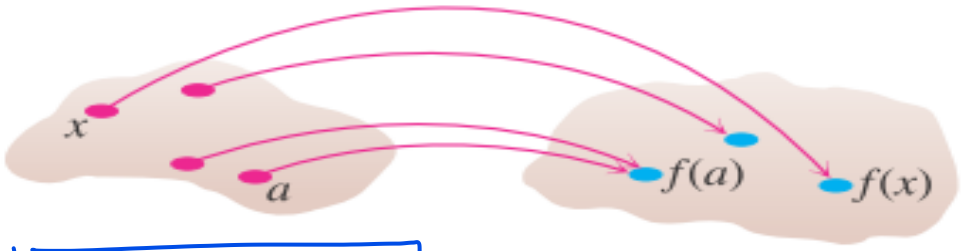
Revision on Functions

- Domain and Range of a Function
- Graphing a Function

Functions; Domain and Range

A function (y = f(x)) is a relation between two variables x and y such that :

for each value of x there exist a unique (single) value of y .



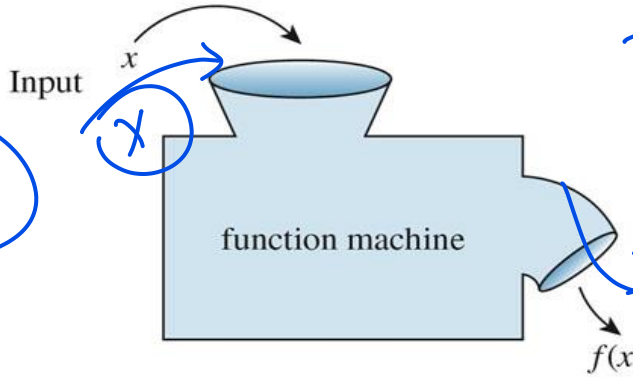
no. That can be as (x)

D = domain set

Y = set containing the range

Re Range of no. That get out of Re function

Function as a machine :



y f(x)

Functions; Domain and Range

Example (function) : Area of square

$$y = x^2 \quad f(x)$$



If x=2 ⟶ y=4 (single value)

If x=3 ⟶ y=9 (single value)

Example (not a function) :

$$y^2 = x$$

$$y = \sqrt{x}$$

not function

$\sqrt{4} = \pm 2$

If x=4 ⟶ y=2 or -2 (multiple values)

Functions; Domain and Range

Domain of the function y=f(x):

The set of all possible values of X

Any value that I can put in X

Range of the function y=f(x):

The set of all values of y when x is the domain

The values that will be output after using x

Symbols of Domain and Range :

Domain of f(x) =D(f) D(f)

Range of f(x)=R(f) R(f)

Functions: Domain and Range

Example **Find the domain and range of the following functions**

a) $y = x^2$

b) $y = \sqrt{x}$

Solution

a) $D(f) = \Re$ $\quad R(f) = [0, \infty)$

b) $D(f) = [0, \infty)$ $\quad R(f) = [0, \infty)$

Functions: Domain and Range

Example Find the domain and range of the following function $y = |x|$

$D(f) = R$

$R(f) = [0, \infty)$

Solution

$$D(f) = \Re \qquad R(f) = [0, \infty)$$

Example: Find the domain of the following function $y = \sqrt{x-1}$

$x \geq 1 \rightarrow D(f) = [1, \infty)$

$R(f) = [0, \infty)$

Solution

$$x - 1 \geq 0 \ , \ x \geq 1$$

$$D(f) = [1, \infty)$$

Functions: Domain and Range

Example: Find the domain of the following function $y = \frac{1}{\sqrt{2-x}}$

$(2-x) > 0$

Solution

$-x > -2$

$x = -1$

$x < 2$

$2 - x > 0$

$(-\infty, 2[$

$-x > -2$

Multiply by (-1)

$x < 2$

$$D(f) = (-\infty, 2)$$

Functions: Domain and Range

Example: Find the domain of the following function $y = \frac{1}{x-2}$

$x - 2 \neq 0$

Solution

$x \neq 2$

$D(f) = R - \{2\}$

$$x - 2 \neq 0$$

$$x \neq 2$$

$$\boldsymbol{D(f) = R - \{2\}}$$

Functions: Domain and Range

Example: Find the domain of the following function $y = \sqrt{4 - x^2}$

Solution

$4 - x^2 \geq 0$

$-x^2 \geq -4$

Multiply by (-1)

$x^2 \leq 4$

$-2 \leq x \leq 2$

$D(f) = [-2, 2]$

$4 - x^2 \geq 0$

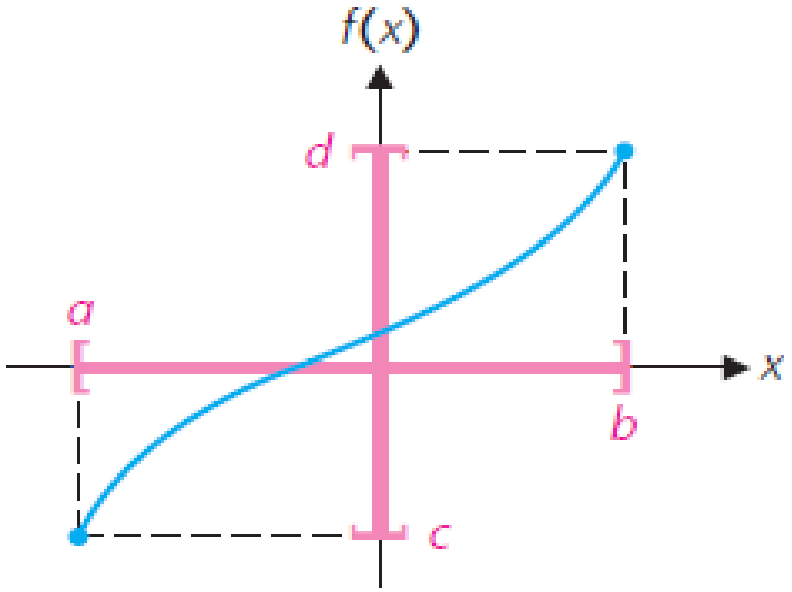
$-x^2 \geq -4$

$x^2 \leq 4$

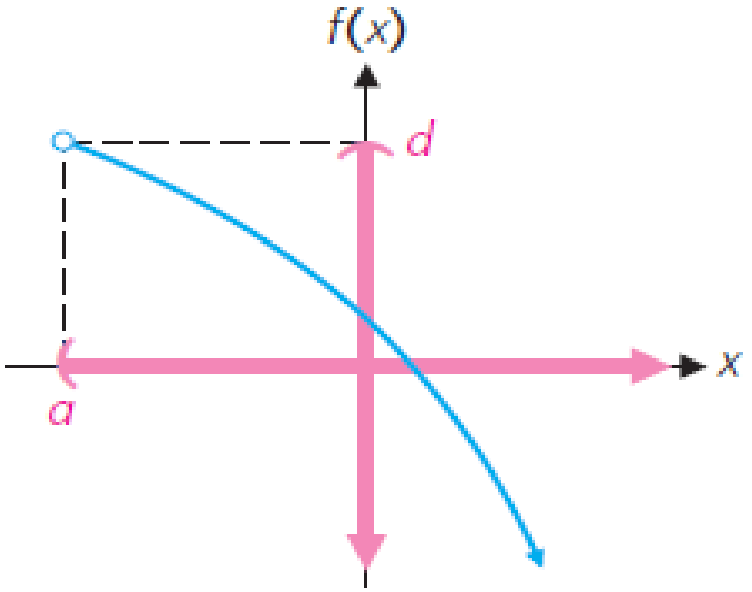
$-2 \leq x \leq 2$

$D(f) = [-2, 2]$

Domain and Range (From Graph)



Domain $f = [a, b]$
Range $f = [c, d]$



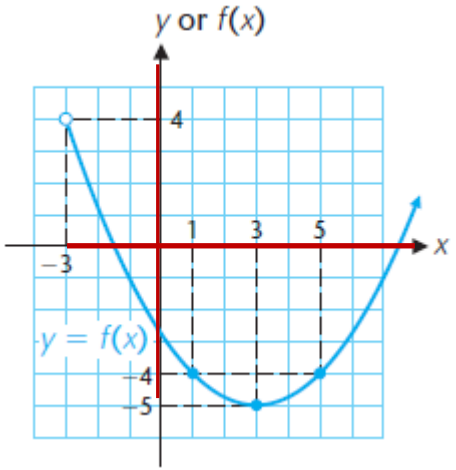
Domain $f = (a, \infty)$
Range $f = (-\infty, d)$

Domain: Look at horizontal axis (from left to right)

Range: Look at vertical axis (down to up)

Domain and Range (From Graph)

Example **Find the domain and range of the following function**



Domain: (from left to right)

Domain: $-3 < x < \infty$ or $(-3, \infty)$

Range: Look at vertical axis (down to up)

Range: $-5 \leq y < \infty$ or $[-5, \infty)$

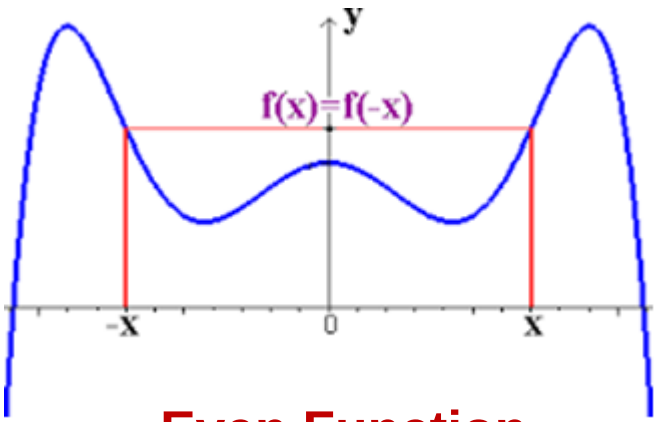
Symmetry: Even and Odd Functions

DEFINITIONS A function $y = f(x)$ is an

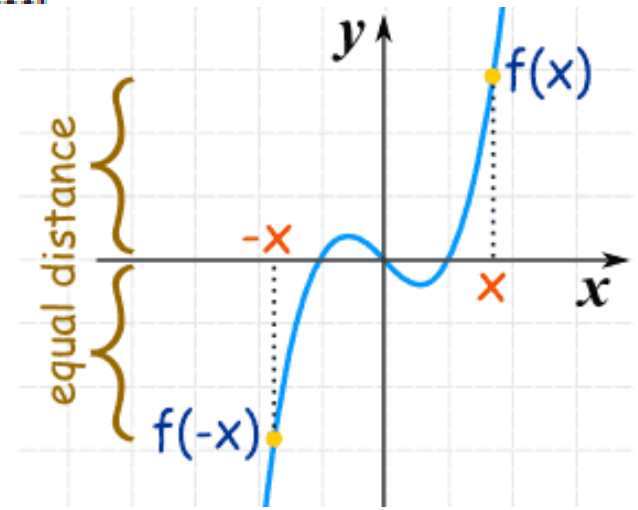
even function of x if $f(-x) = f(x)$,

odd function of x if $f(-x) = -f(x)$,

for every x in the function's domain.



Even Function
Symmetric about y-axis



Odd Function

Symmetry: Even and Odd Functions

To check the Symmetry, replace each x by -x

Examples:

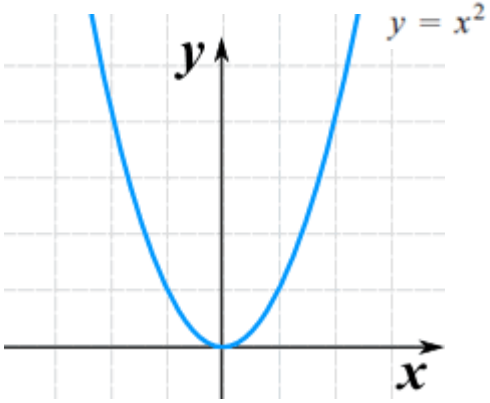
$x^2 + 4x - 3$

i. $f(x) = x^2 + 4|x| - 3$ even

ii. $f(x) = x\sqrt{3 - x^2}$ $-x\sqrt{3-x^2}$ odd

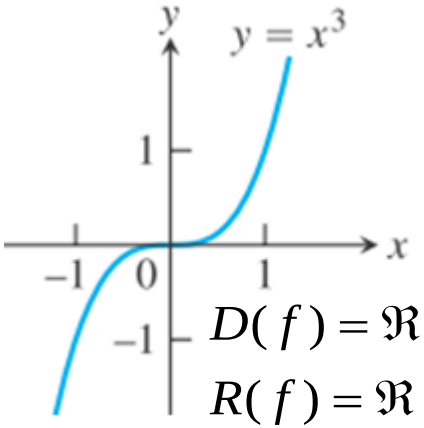
iii. $f(x) = x^3 + 4x^2$ $-x^3 + 4x^2$ niet even noch odd

Graphs of Some Common Functions



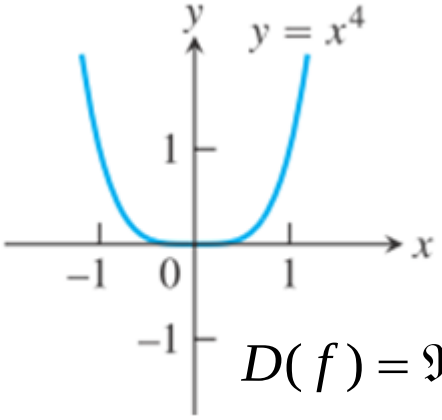
$D(f) = \Re$

$R(f) = [0, \infty)$



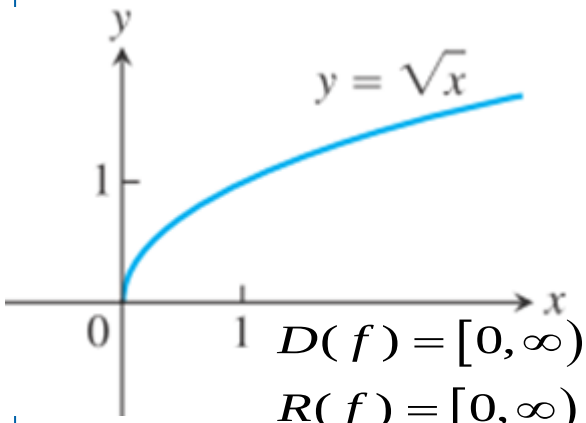
$D(f) = \Re$

$R(f) = \Re$



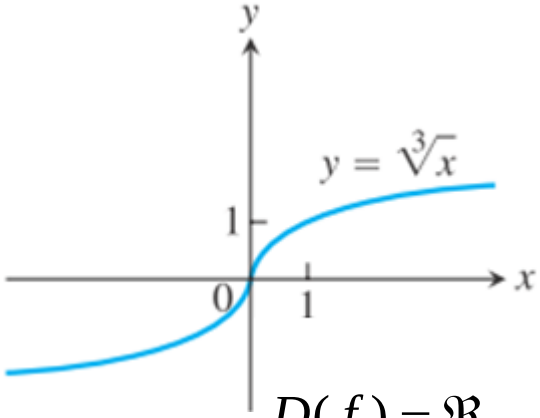
$D(f) = \Re$

$R(f) = [0, \infty)$



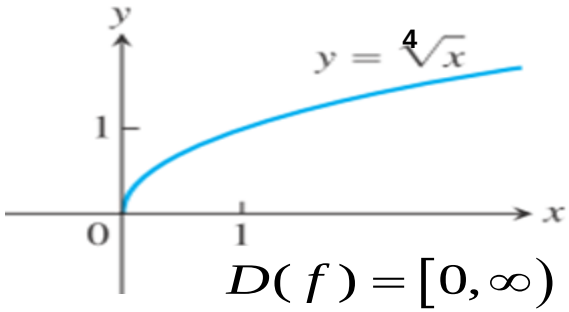
$D(f) = [0, \infty)$

$R(f) = [0, \infty)$



$D(f) = \Re$

$R(f) = \Re$

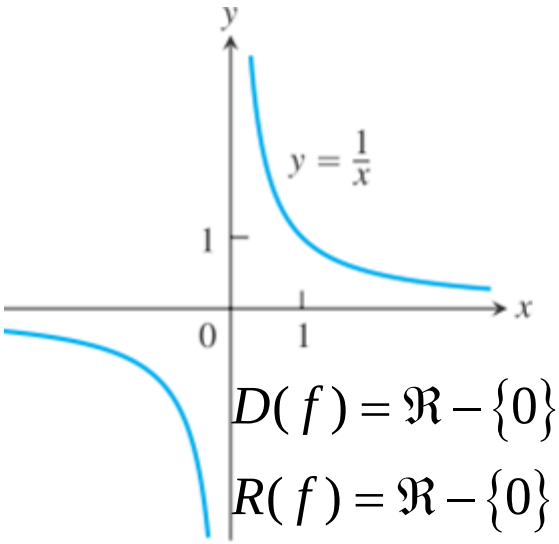


$D(f) = [0, \infty)$

$R(f) = [0, \infty)$

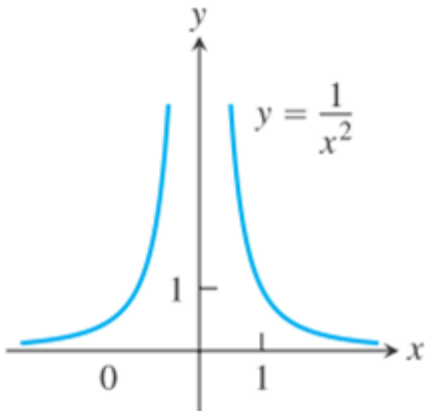
Useful link

Graphs of Some Common Functions



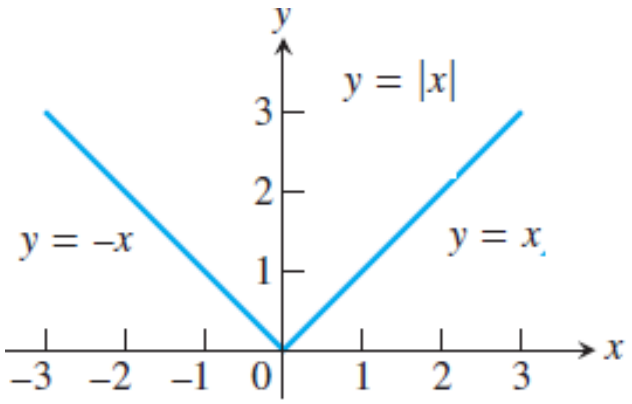
$D(f) = \Re - \{0\}$

$R(f) = \Re - \{0\}$



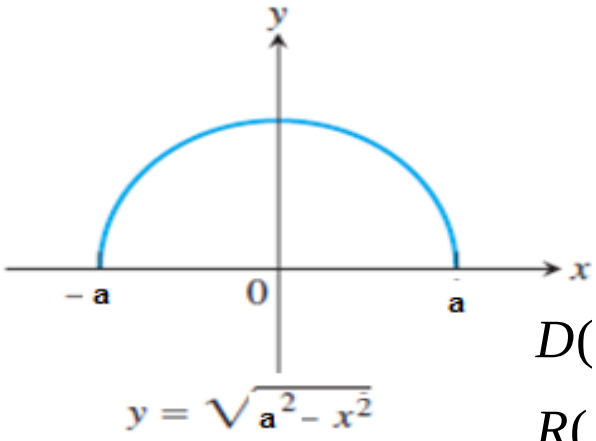
$D(f) = \Re - \{0\}$

$R(f) = (0, \infty)$



$D(f) = \Re$

$R(f) = [0, \infty)$



$D(f) = [-a, a]$

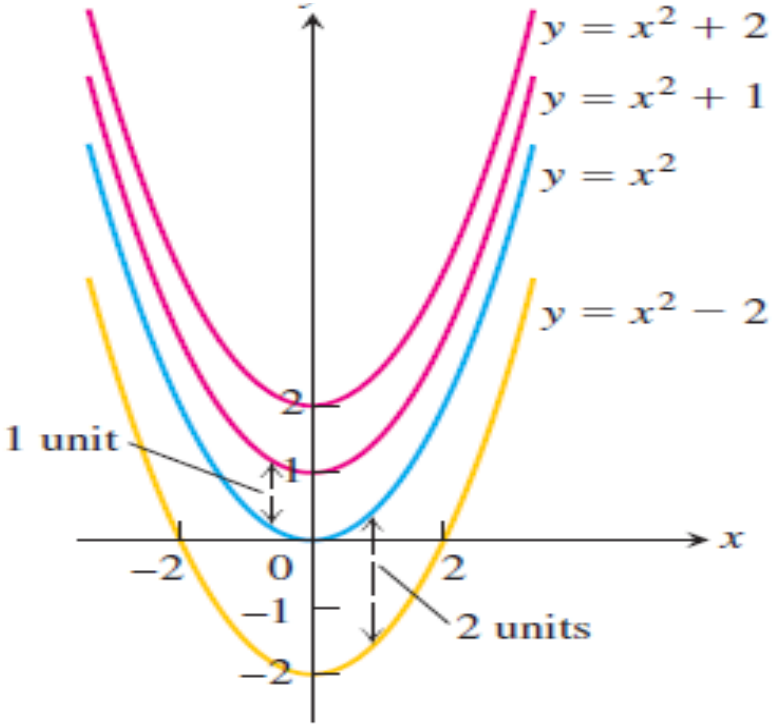
$R(f) = [0, a]$

New functions from old functions
Shifting a Graph of a Function: Vertical Shift

Vertical Shifts

$y = f(x) + k$ Shifts the graph of f *up* k units if $k > 0$

Shifts it *down* $|k|$ units if $k < 0$

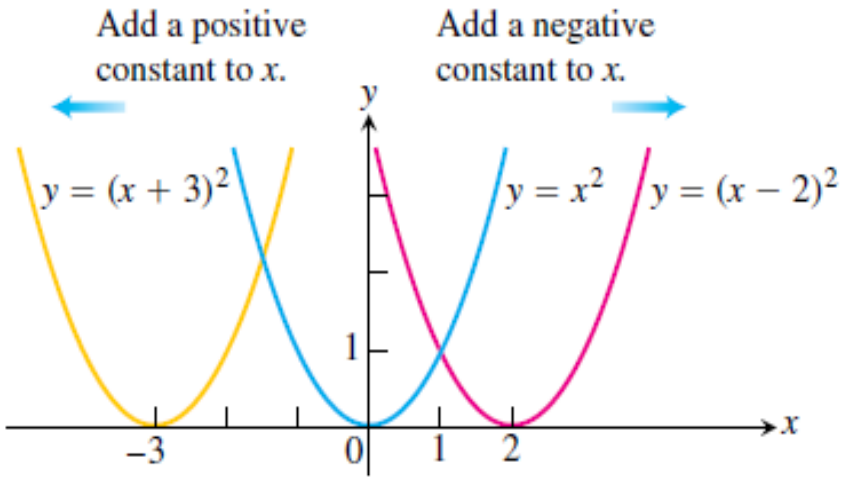


New functions from old functions
Shifting a Graph of a Function: Horizontal Shift

Horizontal Shifts

$y = f(x + h)$ Shifts the graph of f *left* h units if $h > 0$

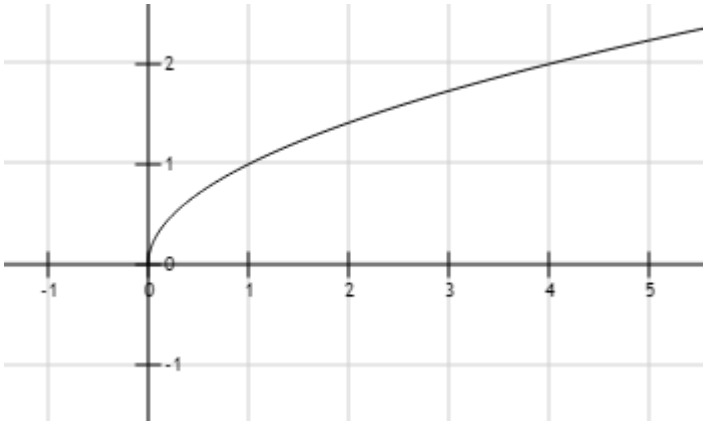
Shifts it *right* $|h|$ units if $h < 0$



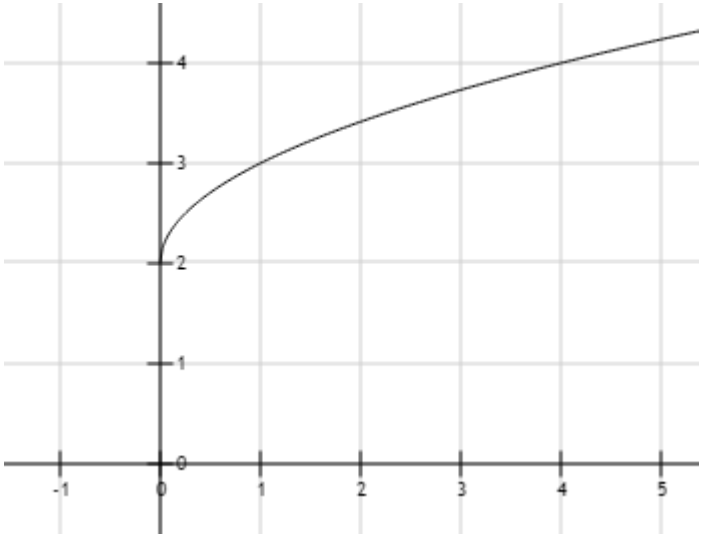
New functions from old functions

Example **Sketch in steps** $y = \sqrt{x}+2$

Step1 $y = \sqrt{x}$



Step2 $y = \sqrt{x}+2$

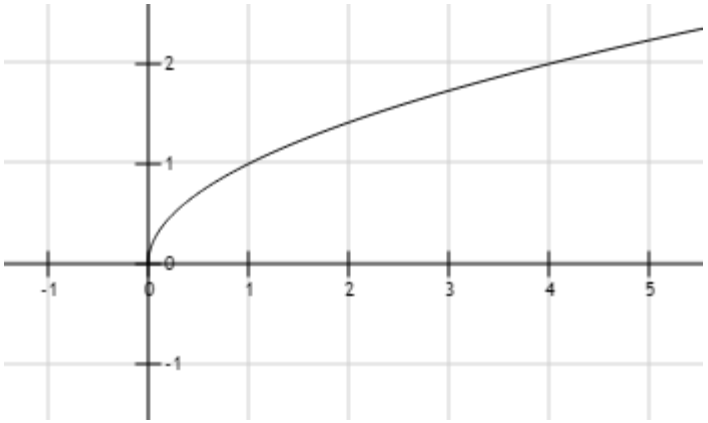


Vertical Shift

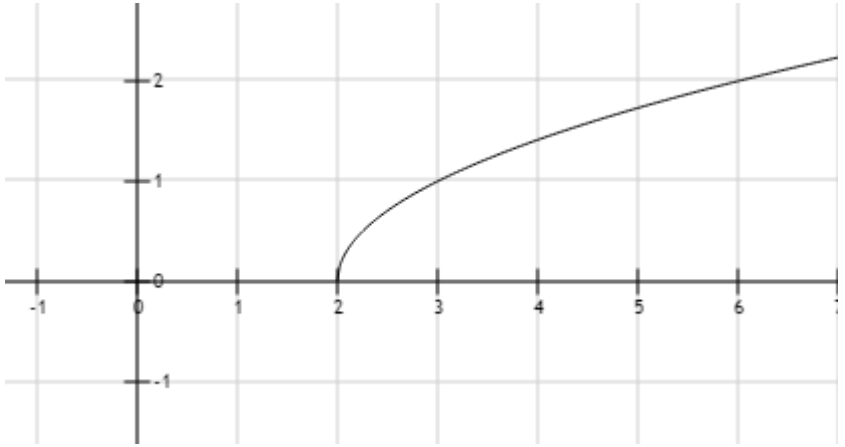
New functions from old functions

Example Sketch in steps $y = \sqrt{x-2}$

Step 1 $y = \sqrt{x}$



Step 2 $y = \sqrt{x-2}$

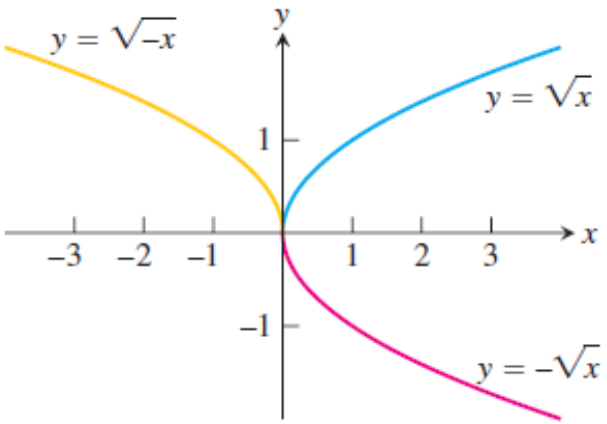


Horizontal Shift

New functions from old functions
Reflecting a Graph of a Function

$y = -f(x)$ Reflects the graph of f across the x-axis.

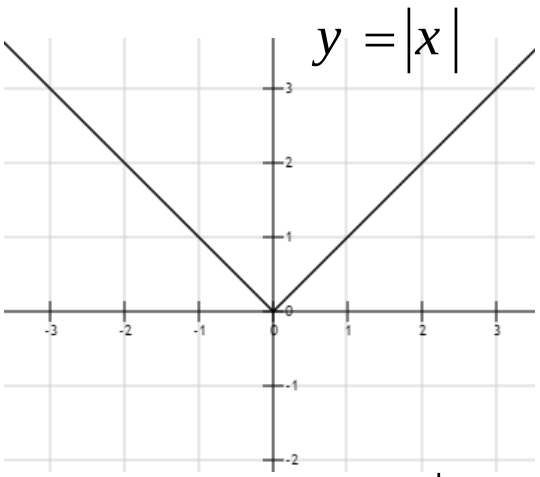
$y = f(-x)$ Reflects the graph of f across the y-axis.



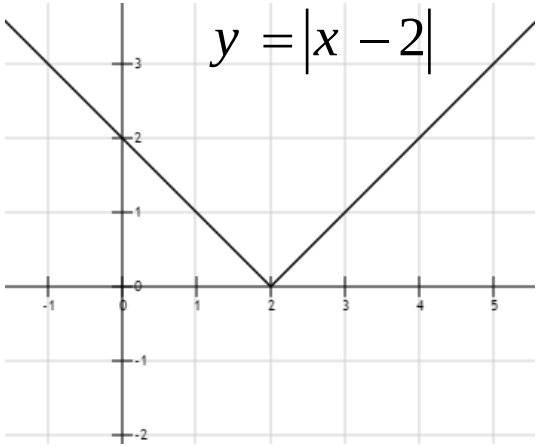
New functions from old functions

Example Sketch in steps $y = 1 - |x - 2|$

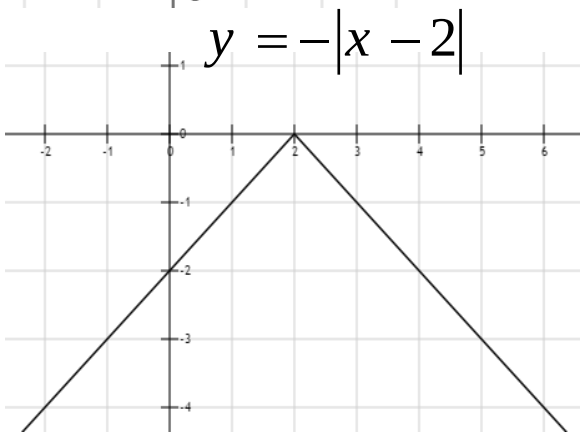
Step 1



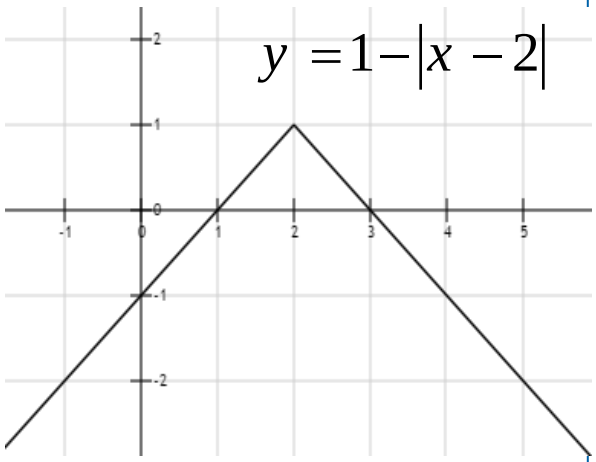
Step 2



Step 3



Step 4



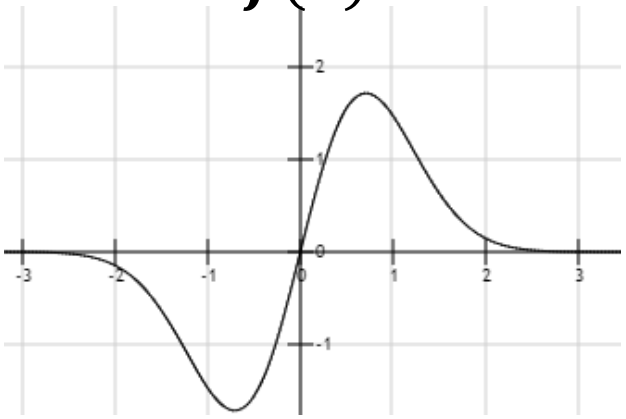
New functions from old functions
Absolute of a Function

How to sketch $|f(x)|$?

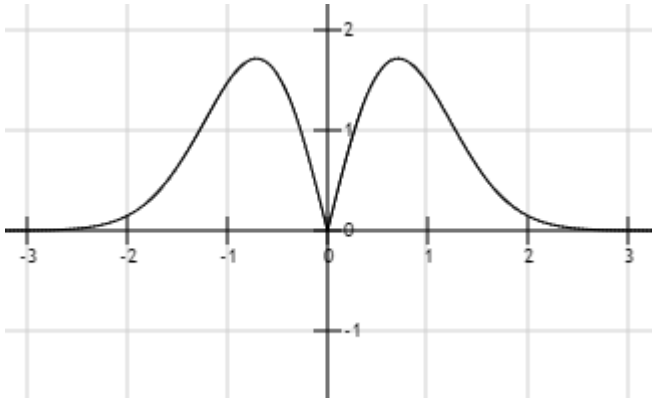
- Sketch $f(x)$
- Negative y-value ⟶ Positive y-value

Example

$f(x)$



$|f(x)|$



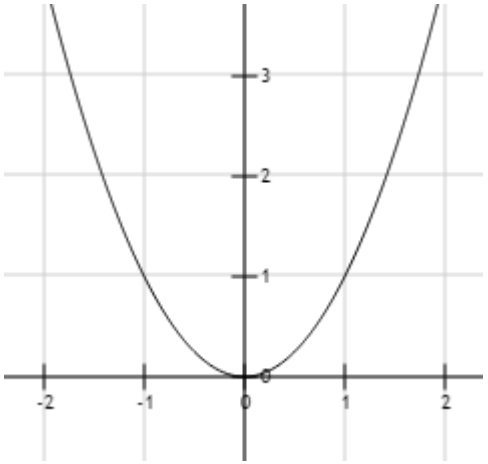
Only the negative part in the graph is reflected across the x-axis

New Functions from Old Functions
Absolute of a Function

Example **Sketch in steps** $y = |x^2 - 4|$

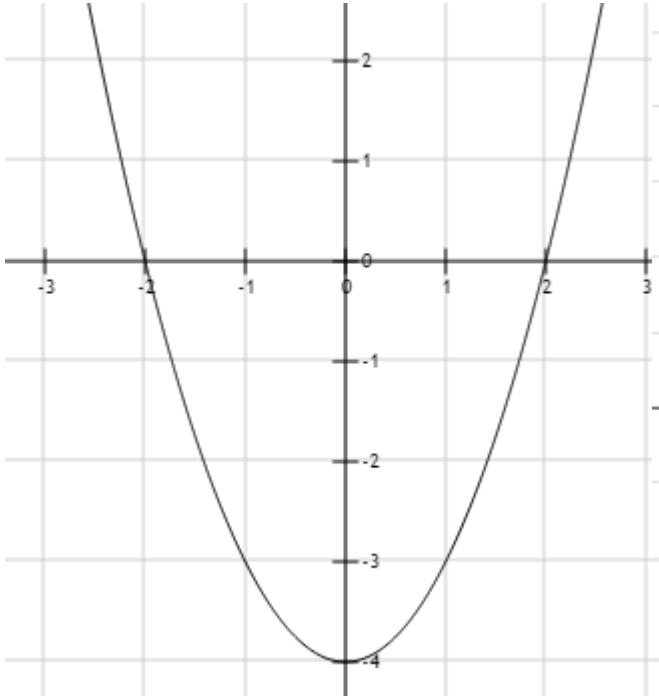
Step 1

$y = x^2$



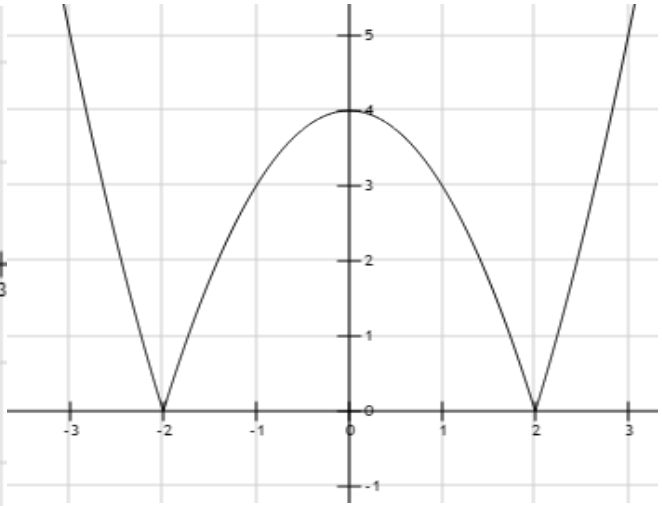
Step 2

$y = x^2 - 4$



Step 3

$y = |x^2 - 4|$

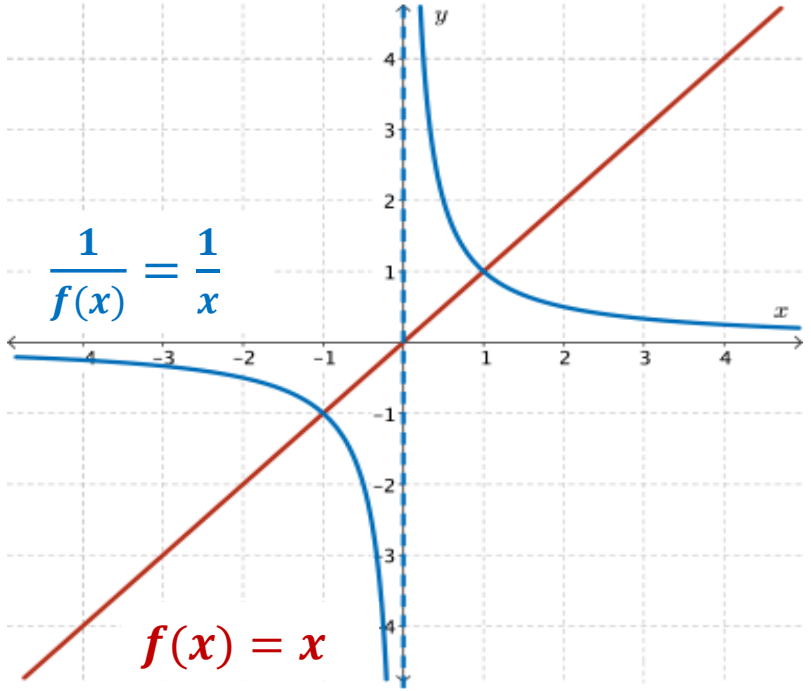


New functions from old functions
Reciprocal of a Function

How to sketch $\frac{1}{f(x)}$?

Example **Sketch $y = \frac{1}{x}$**

- **Sketch the function $f(x) = x$**
- **Sketch the reciprocal $\frac{1}{f(x)} = \frac{1}{x}$**



New functions from old functions
Reciprocal of a Function

Note that:

If $f(x)$ is +ve ⟶ $\frac{1}{f(x)}$ is also +ve

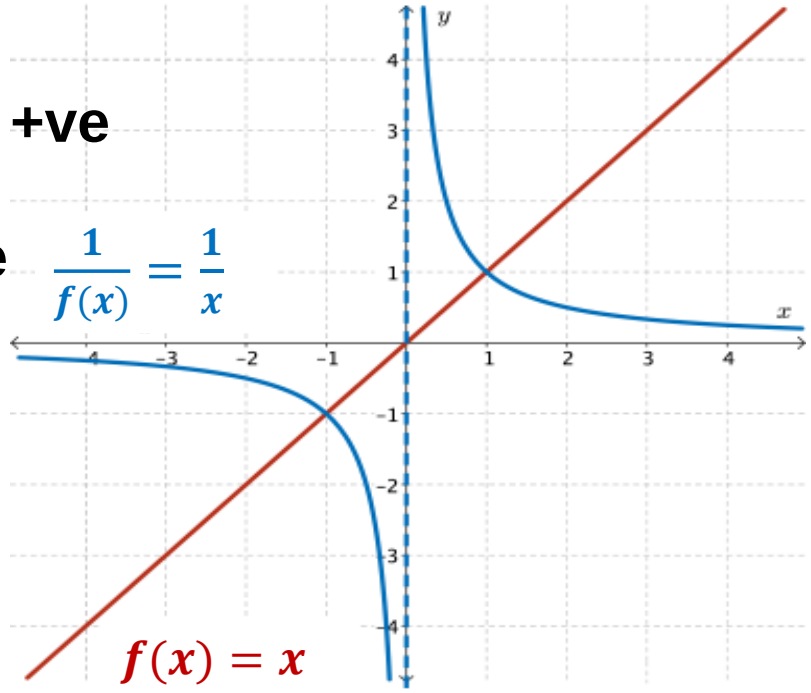
If $f(x)$ is -ve ⟶ $\frac{1}{f(x)}$ is also -ve

If $f(x)$ is large ⟶ $\frac{1}{f(x)}$ is small

If $f(x)$ is small ⟶ $\frac{1}{f(x)}$ is large

If $f(x) \to \pm\infty$ ⟶ $\frac{1}{f(x)} \to$ zero

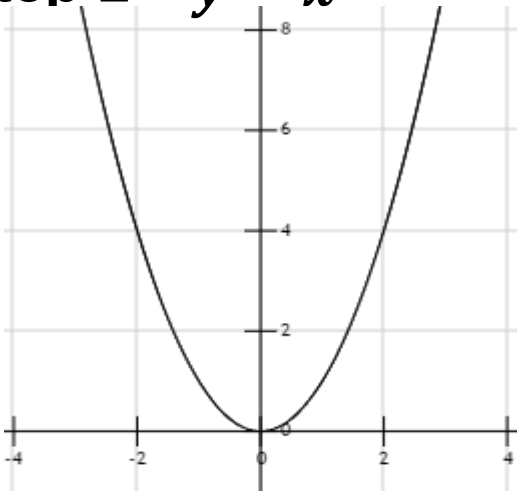
If $f(x) \to$ zero ⟶ $\frac{1}{f(x)} \to \pm\infty$



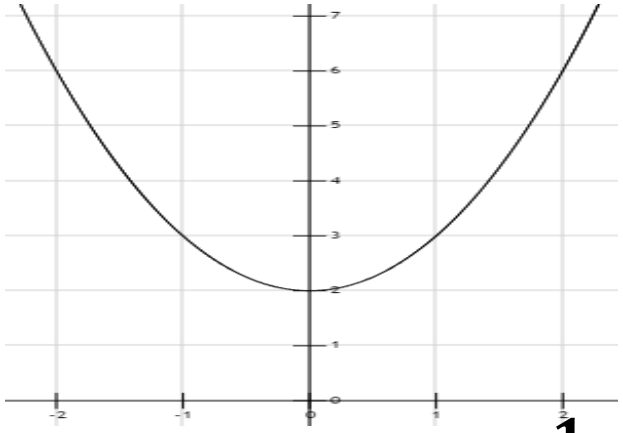
New functions from old functions
Reciprocal of a Function

Example Sketch in steps $f(x) = \frac{1}{x^2+2}$

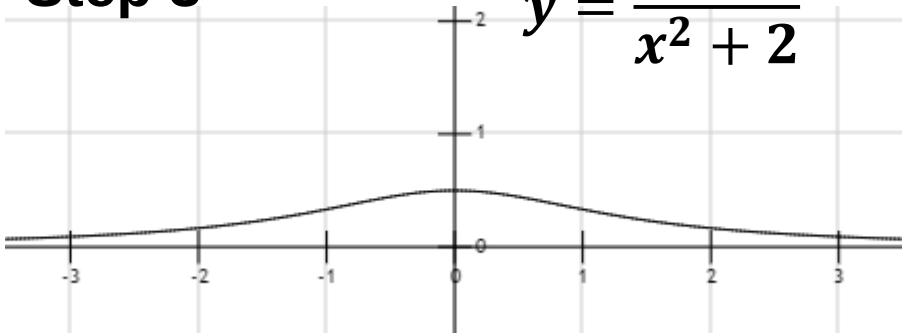
Step 1 $y = x^2$



Step 2 $y = x^2 + 2$



Step 3 $y = \frac{1}{x^2+2}$



New functions from old functions

Example **Sketch in steps the following functions**

a. $y = \frac{1}{\sqrt{x}}$

b. $y = \frac{1}{\sqrt{1-x^2}}$

Try to solve!

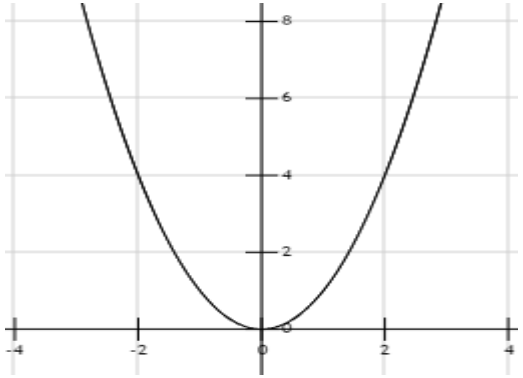
c. $y = \frac{1}{|x|+1}$

d. $y = \frac{1}{2-\sqrt{x}}$

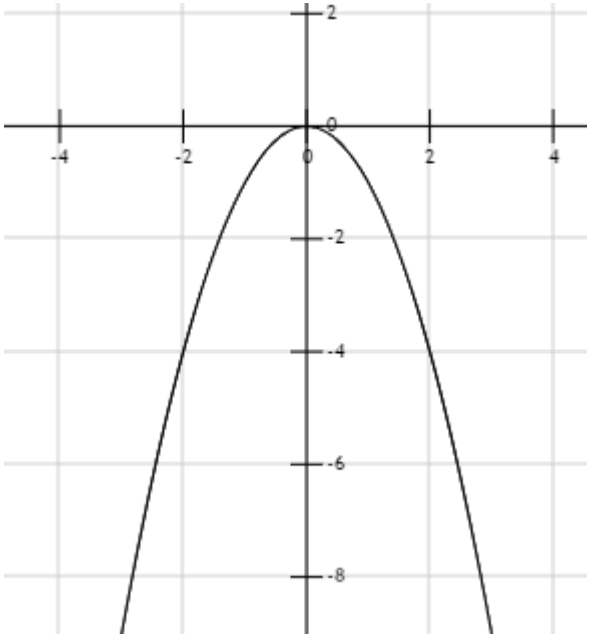
New functions from old functions Example

Example Sketch in steps $f(x) = \frac{1}{4-x^2}$

Step 1 $y = x^2$



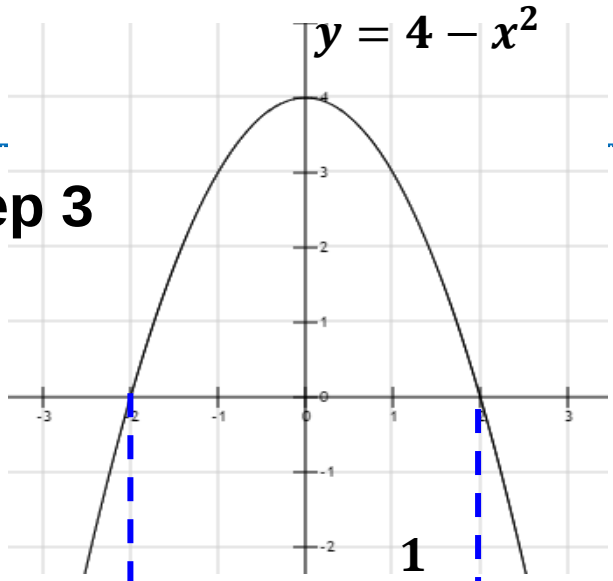
Step 2 $y = -x^2$



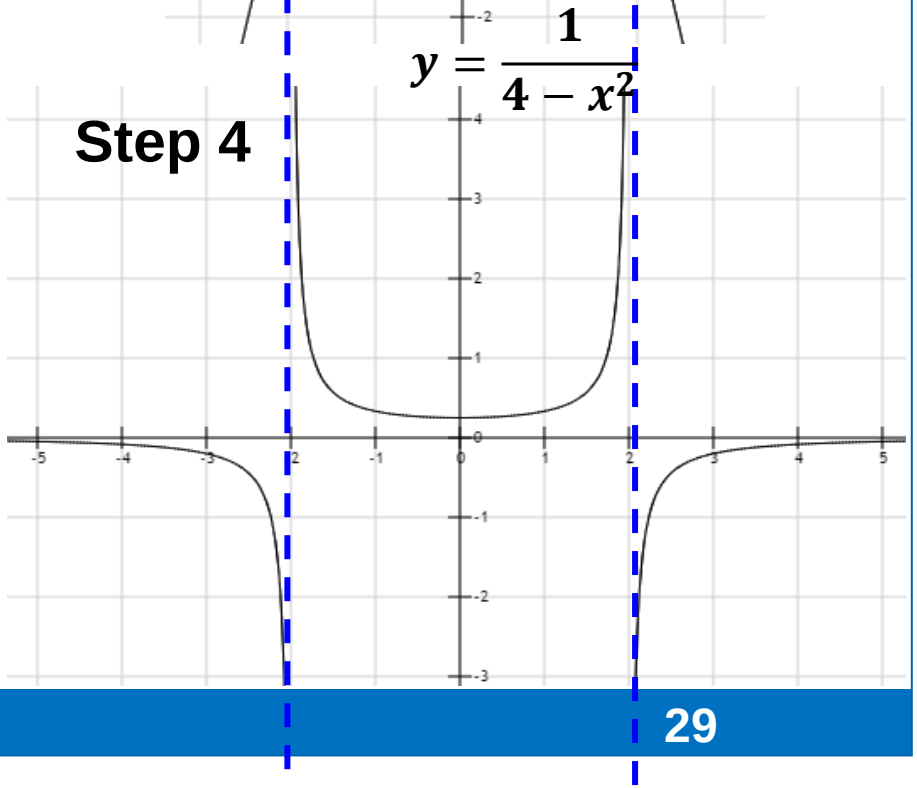
New functions from old functions
Example Continued

Example Sketch in steps $f(x) = \frac{1}{4-x^2}$

Step 3



Step 4





See you next week